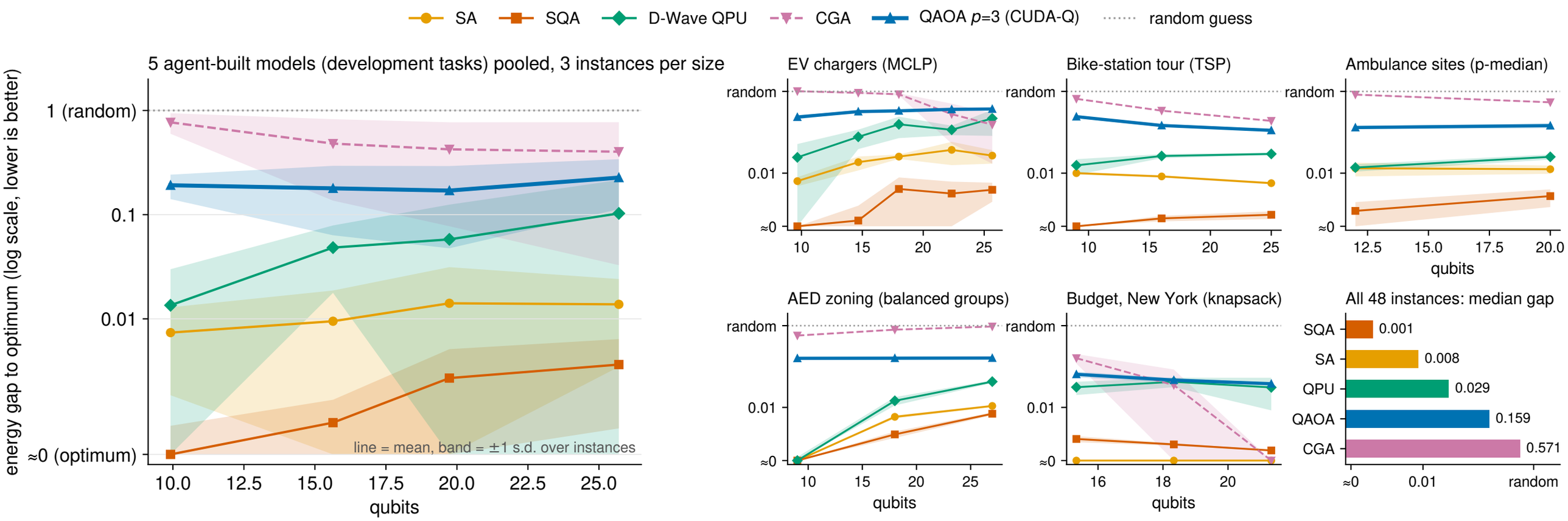
SA
SQA
D-Wave QPU
CGA
QAOA p=3 (CUDA-Q)
random guess
5 agent-built models (development tasks) pooled, 3 instances per size
energy gap to optimum (log scale, lower is better)
1 (random)
0.1
0.01
≈0 (optimum)
line = mean, band = ±1 s.d. over instances
10.0
12.5
15.0
17.5
20.0
22.5
25.0
qubits
EV chargers (MCLP)
Bike-station tour (TSP)
Ambulance sites (p-median)
AED zoning (balanced groups)
Budget, New York (knapsack)
All 48 instances: median gap
random
0.01
≈0
10
15
20
25
12.5
15.0
17.5
20.0
16
18
qubits
SQA
SA
QPU
QAOA
CGA
0.001
0.008
0.029
0.159
0.571